

Tianhe Wang

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RESEARCH PROGRAM

Computational and Cognitive Neuroscience; Motor Control and Motor Learning; Cerebellum and Cognition

PROFESSIONAL EXPERIENCE

2026-present | C.V. Starr Research Fellow
 Princeton Neuroscience Institute, Princeton University, Princeton, NJ, USA

EDUCATION

2020-2026 | PhD in Psychology, concentration in Cognitive Neuroscience
 Department of Psychology, University of California, Berkeley, CA USA
 Advisor: Richard Ivry

2016-2020 | Bachelor of Science in Biological Sciences and Psychology
 School of Life Sciences, Peking University, Beijing, China

PUBLICATIONS

Reviews and Book Chapters (*indicates corresponding author)

In press | **T Wang***, D Narain, RB Ivry
 Timing behind motor control
 In M. Manto, D. Timmann, & L. Clement (Eds.), Roles of the Cerebellum in Motor Control: Foundations, Experimental Approaches and Clinical Applications. Academic Press.

In press | **T Wang***, D Narain, RB Ivry
 Timing and the cerebellum
 In M. Manto et al. (Eds.), Encyclopedic Dictionary of the Cerebellum. Springer.

In prep | **T Wang***, D Whitney, JS Tsay
 Action statistics shape motor planning
 Invited, Trends in Cognitive Sciences

Manuscripts Under Review

2026 | **T Wang***, D Whitney
 Statistical factorization in motor control: Opposing biases from statistics on different timescales enable fast and accurate behavior
 bioRxiv.

2025 | **T Wang***, T Lam, JA Taylor, RB Ivry
 Distinct implicit contributions to action selection and action execution in sensorimotor adaptation
 bioRxiv.

Peer-reviewed Publications

2026 | **T Wang***, R Morehead, A Jiang, RB Ivry, JS Tsay*
 Motor biases reflect a misalignment between visual and proprioceptive reference frames
 eLife

2026 | L Dowdall et al. (including **T Wang**)

- Integrated approaches for investigating the neural bases of movement
Journal of Neurophysiology
- 2025 **T Wang***, RB Ivry
Contextual effects during sensorimotor adaptation are an emergent property of population coding in a cerebellar-inspired model
Science Advances (Featured in social media highlights)
- 2025 **T Wang***, J Li, RB Ivry
Attention defines the context for implicit sensorimotor adaptation
Journal of Neuroscience
- 2025 **T Wang***, Y Fang, D Whitney
Sequential effects in reaching reveal efficient coding in motor planning
Science Advances
- 2024 **T Wang***, G Avraham, JS Tsay, SJ Abram, RB Ivry
Perturbation variability does not influence implicit sensorimotor adaptation
PLOS Computational Biology
- 2024 **T Wang***, G Avraham, JS Tsay, T Thummala, RB Ivry
Advanced feedback enhances sensorimotor adaptation
Current Biology
- 2023 S Chen, **T Wang**, Y Bao*
Serial dependence in timing at the perceptual level being modulated by working memory
PsyCh Journal
- 2023 NM van Mastrigt*, JS Tsay, **T Wang**, G Avraham, SJ Abram, K van der Kooij, JBJ Smeets, RB Ivry
Implicit reward-based motor learning
Experimental Brain Research
- 2023 **T Wang***, Y Luo, RB Ivry, JS Tsay, E Pöppel, Y Bao*
A unitary mechanism underlies adaptation to both local and global environmental statistics in time perception
PLOS Computational Biology
- 2022 JS Tsay*, T Najafi, L Schuck, **T Wang**, RB Ivry
Implicit sensorimotor adaptation is preserved in Parkinson's disease
Brain Communications
- 2021 **T Wang***, JA Taylor
Implicit adaptation to mirror reversal is in the correct coordinate system but the wrong direction
Journal of Neurophysiology
- 2021 **T Wang**, Z Zhu, I Kana, Y Yu, H He, K Wei*
Accuracy of hand localization is subject-specific and improved without performance feedback
Scientific Reports

FELLOWSHIPS & FUNDS

2026-2029	C.V. Starr Fellowship, Princeton University & C.V. Starr Foundation
2025	Travel Grant, Cerebellum Gordon Research Conference
2025	NCM Scholarship Award, Society for the Neural Control of Movement
2022	Travel Fellowship, University of California, Berkeley
2021	Department Fellowship, University of California, Berkeley
2020	Distinguished Undergraduate Scholarship, Peking University

TALKS

2026	Statistical Regulators in Motor Learning and Motor Control Wu Tsai Institute Postdoctoral Symposium, New Haven, USA
2025	Contextual Effects During Implicit Recalibration Are an Emergent Property of Population Coding in a Cerebellar Model Society for the Neural Control of Movement, Panama City, Panama
2025	Cerebellar Ataxia Patients Exhibit Accurate Representation of a Temporal Prior Despite Noisy Evidence Cerebellum Gordon Research Seminar, Vaud, Switzerland
2023	A Cerebellar Population Coding Model for Sensorimotor Adaptation 2023 Berkeley Neuroscience Conference, Tahoe City, CA, USA
2023	A Cerebellar Population Coding Model for Sensorimotor Adaptation Advances in Motor Learning and Motor Control, Washington, DC, USA
2022	Serial Dependence Reveals an Active Suppression in the Sequential Motor Planning Advances in Motor Learning and Motor Control, San Diego, CA, USA
2021	Why is Online Feedback More Effective than Endpoint Feedback for Sensorimotor Adaptation? Advances in Motor Learning and Motor Control, Virtual Conference
2021	Sequential Effect in Duration Reproduction Is from Both Perception and Response International Conference on Spatial Cognition, Rome, Italy

TEACHING & SERVICE

Teaching Experience

2024 & 2025	Instructor Neuroscience Bootcamp Department of Neuroscience, UC Berkeley
2023	Graduate Student Instructor Human Neuroanatomy Department of Psychology, UC Berkeley
2021	Graduate Student Instructor Cognitive Neuroscience Department of Psychology, UC Berkeley

2018	Undergraduate Teaching Instructor Methods in Psychology Research Department of Psychology, Peking University
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Service Experience

2027	Co-Chair, From Movement to Mind: Rethinking Cerebellar Computation, Satellite Meeting, Society for the Neural Control of Movement, Seville, Spain
2026-2028	Board Member, Society for the Neural Control of Movement
2025	Graduate Student Mentor, Bioengineering Scholars Program, UC Berkeley
2023-2024	Faculty Search Committee, Biological Basis of Behavior, UC Berkeley
2023	Graduate Student Mentor, Research Experience Pathways in Psychology Program, UC Berkeley

Ad Hoc Reviewer

Nature Reviews Psychology; Nature Communications; Journal of Neuroscience; Communications Biology; Communications Psychology; BMC Biology; Journal of Neurophysiology; Cerebral Cortex; eNeuro; npj Science of Learning; The European Journal of Neuroscience; Scientific Reports; Human Movement Science; Journal of Behavioral and Brain Sciences; PLOS ONE

ADVISING & MENTORSHIP

Undergraduate Research Assistants (*indicates honors thesis student)

2025-present	Suqing Xie - UC Berkeley
2025-present	Clinton D Nguyen - UC Berkeley
2025-2026	Sara Sotelo - UC Berkeley
2025	Noah A Math - UC Berkeley
2025	Mihail Dimitrov - UC Berkeley
2025	Ijin Yu - UC Berkeley
2024-2026	Mary Swanson-Wunder - UC Berkeley
2024-2025	Maoxin M Xia* - Wuhan University (Now Ph.D. student at CMU)
2024-2025	Amber Jiang - UC Berkeley (Now M.S. student at UCLA)
2023-2025	Yuling Li - UC Berkeley
2023	Marina J Armstrong - UC Berkeley
2022-2024	Tony Lam - UC Berkeley (Now M.D. student at UCLA)
2022-2024	Yifan Fang - UC Berkeley (Now Ph.D. student at Dartmouth College)
2021-2023	Jialin Li - UC Berkeley (Now Ph.D. student at Washington University)
2021-2023	Tanvi Thummala* - UC Berkeley